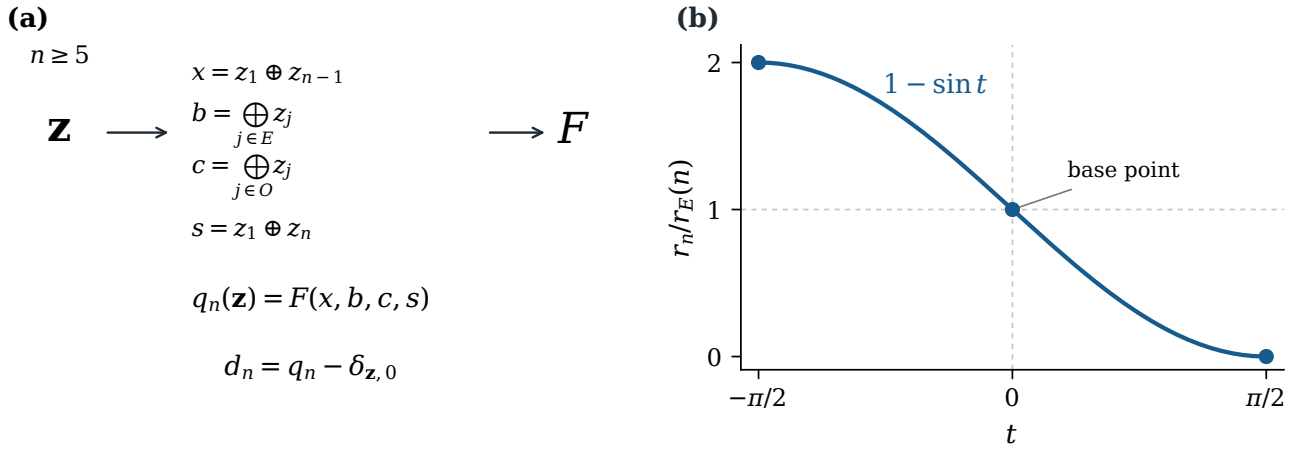
(a)
$n \geq 5$
$\mathbf{z}$
$x = z_1 \oplus z_{n-1}$
$b = \bigoplus_{j \in E} z_j$
$c = \bigoplus_{j \in O} z_j$
$s = z_1 \oplus z_n$
F
$q_n(\mathbf{z}) = F(x, b, c, s)$
$d_n = q_n - \delta_{\mathbf{z},0}$
(b)
$1 - \sin t$
base point
$r_n/r_E(n)$
2
1
0
$-\pi/2$
0
$\pi/2$
t